A PROPERTY OF A REPRESENTABLE TRANSFORMATION DESCENDS ALONG A SURJECTIVE FPPF MAP

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement and the proof each contain an xymatrix display; both are inside the counted spans and are preserved verbatim, with the chapter's own Xy-pic setup retained in the preamble.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-spaces/retained-preamble.tex, sliced mechanically from the chapter's own preamble.tex.

Allowed standard prerequisites (editorial).

- Base change of a representable transformation is representable. Lemma lemma-base-change-representable-transformations, used at proof lines 839-841. Makes a' representable.
- Properties of the relative-representable-property type are local on the base. Definition definition-relative-representable-property, used at proof lines 854-857. Descends property P from the covering to U.

Source reference. Original definition 025V: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2828–2828. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02WB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3798–3798. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme contained in Sch_{fppf}. Let $F, G, H : (Sch/S)^{opp}_{fppf} \to Sets$. Let $\mathcal{P}$ be a property as in Definition 025V. Let $a : F \to G$ be a representable transformation of functors. Let $b : H \to G$ be any transformation of functors. Consider the fibre product diagram*

$$\begin{array}{ccc} H \times_{b,G,a} F & \xrightarrow{b'} & F \\ {\scriptstyle a'}\downarrow & & \downarrow{\scriptstyle a} \\ H & \xrightarrow{b} & G \end{array}$$

Assume that b induces a surjective map of fppf sheaves $H^\# \to G^\#$. In this case, if a' has property $\mathcal{P}$, then also a has property $\mathcal{P}$.

Date: stacks-project:spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. First we remark that by Lemma 02WB the transformation a' is representable. Let $U \in \mathrm{Ob}((Sch/S)_{fppf})$, and let $\xi \in G(U)$. By assumption there exists an fppf covering $\{U_i \to U\}_{i \in I}$ and elements $\xi_i \in H(U_i)$ mapping to $\xi|_U$ via b. From general category theory it follows that for each i we have a fibre product diagram

$$\begin{array}{ccc} U_i \times_{\xi_i, H, a'} (H \times_{b,G,a} F) & \longrightarrow & U \times_{\xi, G, a} F \\ \downarrow & & \downarrow \\ U_i & \longrightarrow & U \end{array}$$

By assumption the left vertical arrow is a morphism of schemes which has property $\mathcal{P}$. Since $\mathcal{P}$ is local in the fppf topology this implies that also the right vertical arrow has property $\mathcal{P}$ as desired. $\square$